

NKPC with Gibbs Sampling

Data:

- US quarterly data.
- Inflation: CPI
- Output gap (1): $GDPC1$ (BN) = $100 \times \text{Deviation of } \ln(GDPC1)$
- Output gap (2): Unemployment gap = Unemployment Rate – Natural Rate of Unemployment
- Output gap (3): Inverse of the markup (BN)
- Output gap (4): Inverse of the markup (original series)
- Competition measure changes by section.
- Inflation Expectations: SPF, 1-year ahead CPI forecast.
- Gibbs run: $n_{iter} = 12000$, burn-in = 4000, thin = 5.

$$\begin{aligned}\pi_t &= \alpha\pi_{t-1} + (1 - \alpha)\mathbb{E}_t\pi_{t+1} + \kappa x_t + e_t \\ x_t &= \phi_1 x_{t-1} + \zeta_t\end{aligned}$$